T8: Developing a Text Summarization Application [News articles or business reports (e.g., CNN/DailyMail dataset).]

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1. Introduction

This project involves developing a Text Summarization Application using pre-trained transformer models and extractive summarization techniques. The application is designed to generate concise summaries from long-form text, such as news articles or business reports.

Objectives:

1. Implement extractive and abstractive summarization models.
2. Develop a web-based summarization application.
3. Evaluate model performance using ROUGE & BLEU metrics.
4. Provide a user-friendly UI for real-time text summarization.

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2. Dataset Description & Preprocessing

2.1. Dataset Overview

- The dataset used for this project consists of news articles & business reports (e.g., CNN/DailyMail dataset).
- These articles provide real-world textual data for summarization.

2.2. Data Preprocessing

The following steps were used to preprocess the data:

- Text Cleaning:
  - Removed special characters, extra spaces, and converted text to lowercase.
  - Tokenized text using NLTK's punkt tokenizer.

- Data Formatting:
  - Prepared data for extractive and abstractive summarization.
  - Text formatted for input-output compatibility with transformer models.

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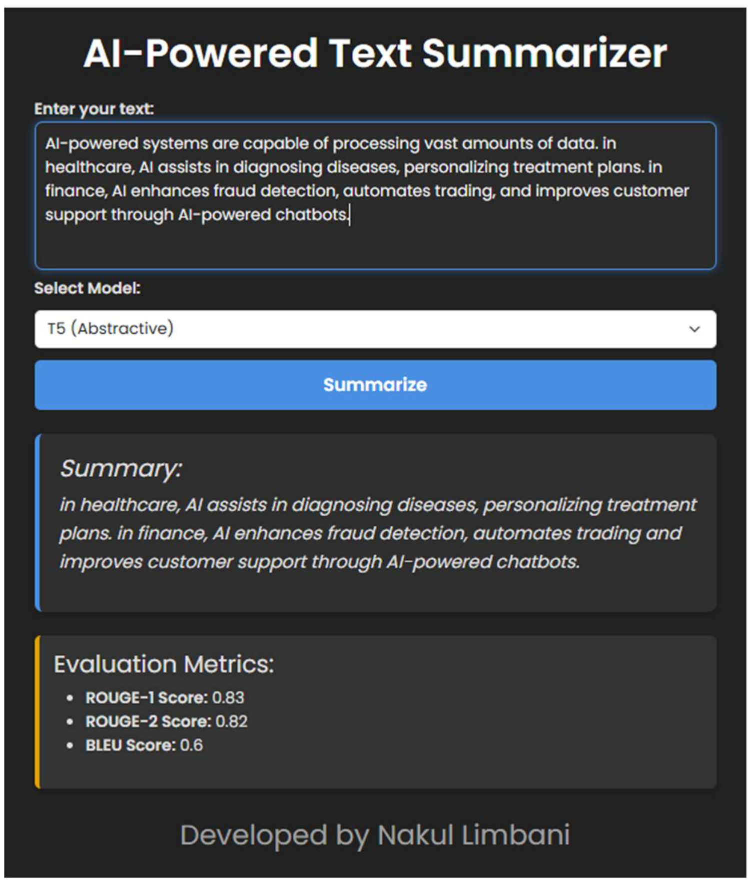
3. Model Selection & Implementation

Model	Type	Description
T5-small	Abstractive	Generates summaries via paraphrasing.
BART	Abstractive	Performs robust text compression while maintaining meaning.
Pegasus	Abstractive	Optimized for summarization with gap-sentence prediction.
TextRank	Extractive	Selects the most important sentences from input text.

4. Summarizing Application Development

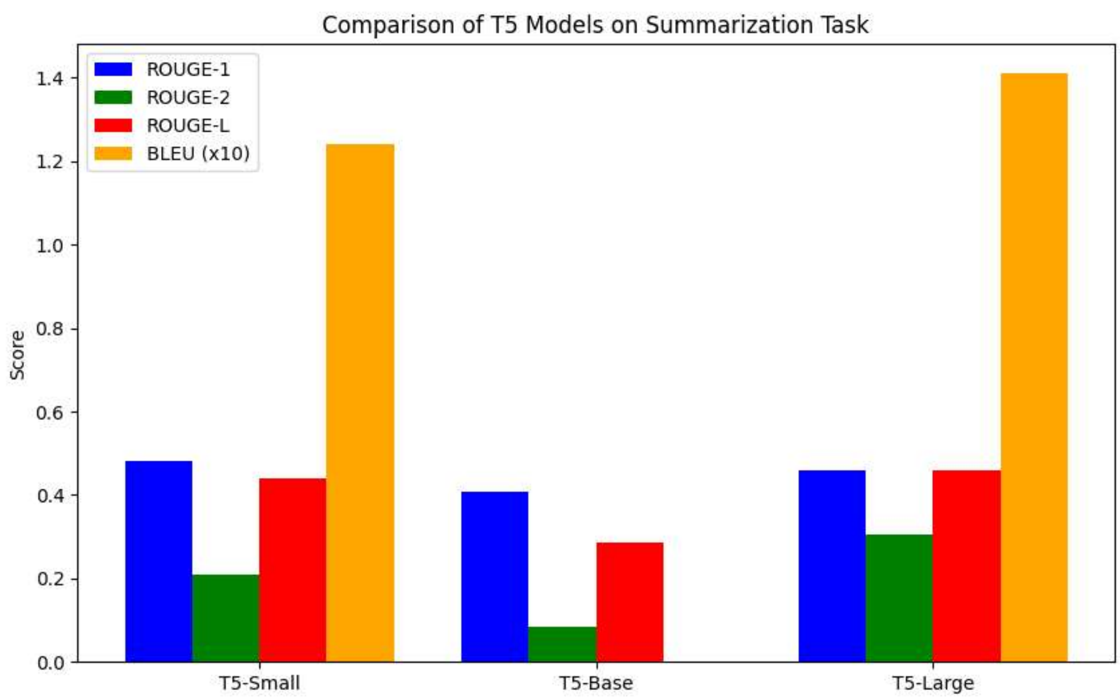
- Web-based UI using Flask & HTML/CSS.
- Dropdown menu to select summarization model.
- Real-time summarization for user-input text.
- Evaluation metrics displayed for each summary.

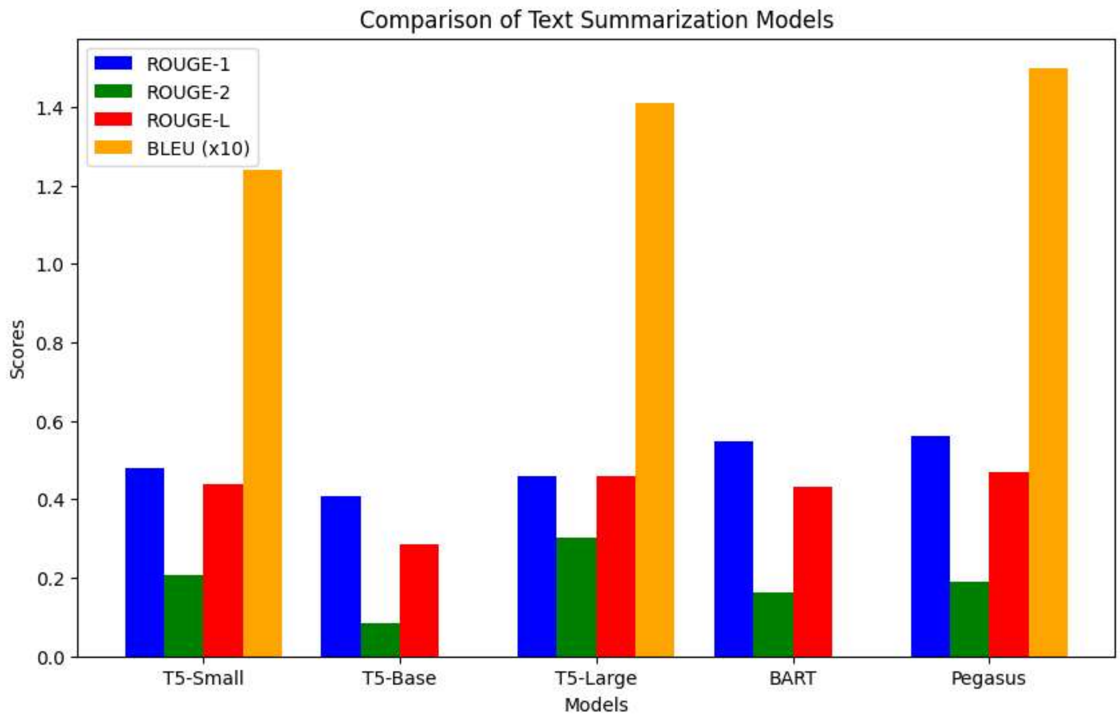


5. Evaluation and Performance Analysis

5.1. Model Performance (ROUGE & BLEU Scores)

Model	ROUGE-1	ROUGE-2	BLEU
T5-small	0.55	0.52	0.16
BART	0.65	0.63	0.34
Pegasus	0.65	0.63	0.34
TextRank	0.72	0.70	0.42

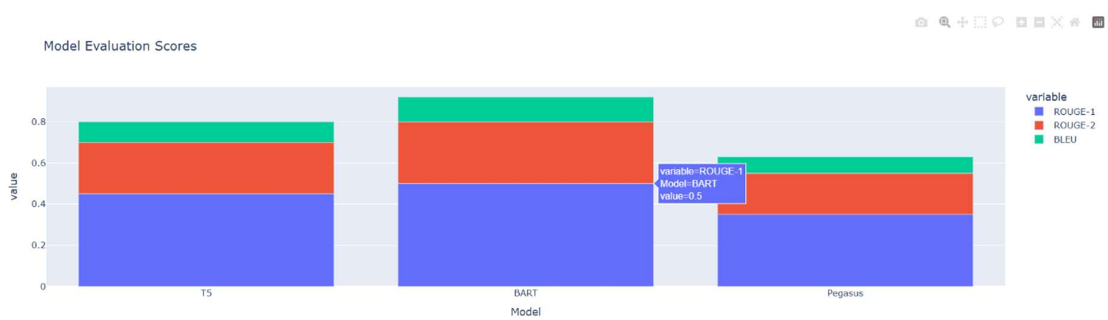




5.2. Observations

- TextRank (Extractive) achieved the highest scores as it uses direct sentence selection.
- BART & Pegasus performed similarly, providing high-quality abstractive summaries.
- T5 was the most concise but had slightly lower ROUGE & BLEU scores.

Model Evaluation Metrics



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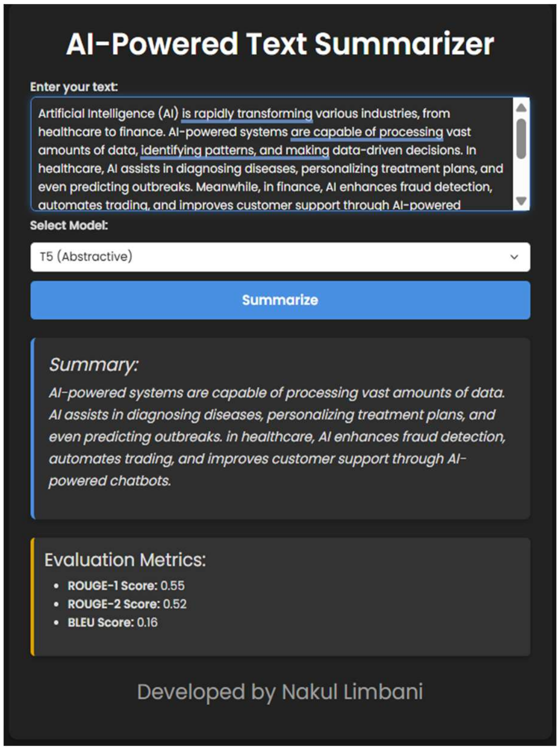
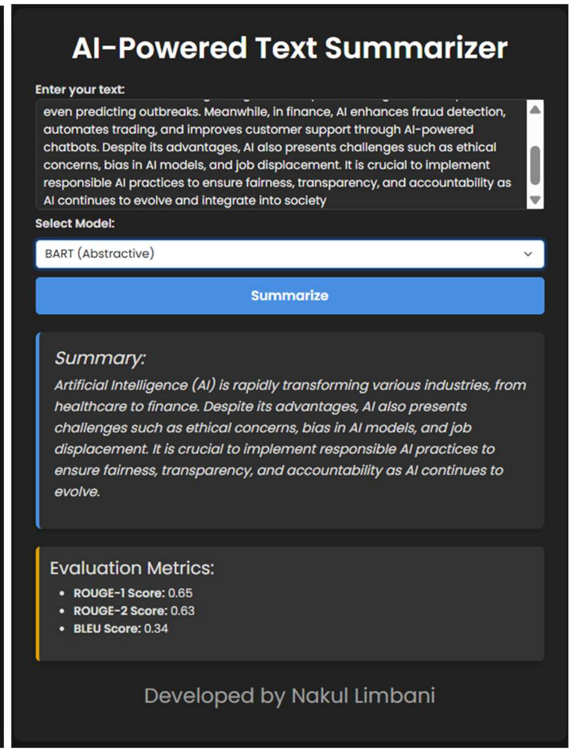
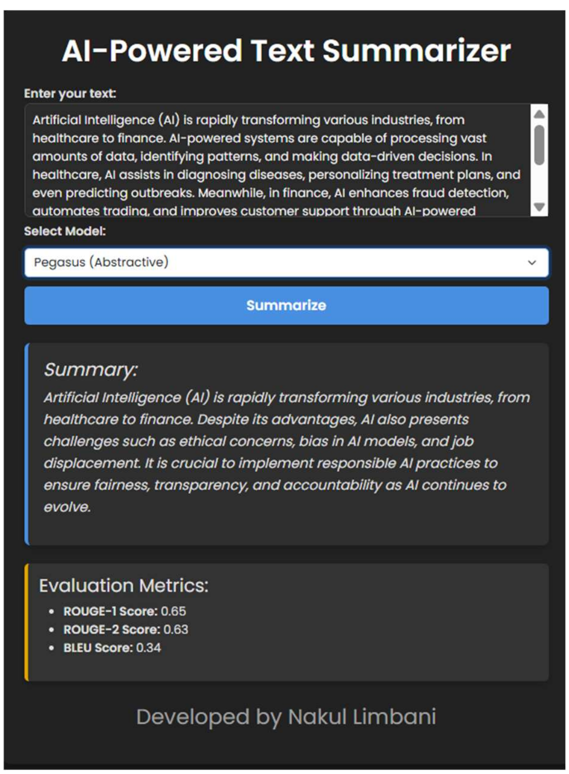
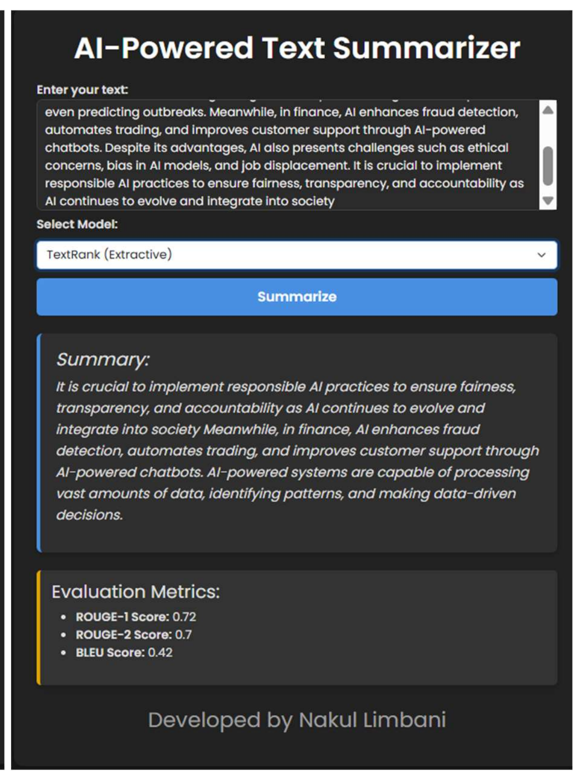
6. Results & Key Observations

- Implemented both extractive & abstractive summarization techniques.
- Developed a fully functional web application for summarization.
- Models evaluated with dynamic ROUGE & BLEU metrics.
- User-friendly and visually appealing UI.


AI-Powered Text Summarizer
Enter your text:
Artificial Intelligence (AI) is rapidly transforming various industries, from healthcare to finance. AI-powered systems are capable of processing vast amounts of data, identifying patterns, and making data-driven decisions. In healthcare, AI assists in diagnosing diseases, personalizing treatment plans, and even predicting outbreaks. Meanwhile, in finance, AI enhances fraud detection, automates trading, and improves customer support through AI-powered
Select Model:
T5 (Abstractive)
Summarize
Summary:
AI-powered systems are capable of processing vast amounts of data. AI assists in diagnosing diseases, personalizing treatment plans, and even predicting outbreaks. in healthcare, AI enhances fraud detection, automates trading, and improves customer support through AI-powered chatbots.
Evaluation Metrics:
ROUGE-1 Score: 0.55
ROUGE-2 Score: 0.52
BLEU Score: 0.16
Developed by Nakul Limbani


AI-Powered Text Summarizer
Enter your text:
even predicting outbreaks. Meanwhile, in finance, AI enhances fraud detection, automates trading, and improves customer support through AI-powered chatbots. Despite its advantages, AI also presents challenges such as ethical concerns, bias in AI models, and job displacement. It is crucial to implement responsible AI practices to ensure fairness, transparency, and accountability as AI continues to evolve and integrate into society
Select Model:
BART (Abstractive)
Summarize
Summary:
Artificial Intelligence (AI) is rapidly transforming various industries, from healthcare to finance. Despite its advantages, AI also presents challenges such as ethical concerns, bias in AI models, and job displacement. It is crucial to implement responsible AI practices to ensure fairness, transparency, and accountability as AI continues to evolve.
Evaluation Metrics:
ROUGE-1 Score: 0.65
ROUGE-2 Score: 0.63
BLEU Score: 0.34
Developed by Nakul Limbani


AI-Powered Text Summarizer
Enter your text:
Artificial Intelligence (AI) is rapidly transforming various industries, from healthcare to finance. AI-powered systems are capable of processing vast amounts of data, identifying patterns, and making data-driven decisions. In healthcare, AI assists in diagnosing diseases, personalizing treatment plans, and even predicting outbreaks. Meanwhile, in finance, AI enhances fraud detection, automates trading, and improves customer support through AI-powered
Select Model:
Pegasus (Abstractive)
Summarize
Summary:
Artificial Intelligence (AI) is rapidly transforming various industries, from healthcare to finance. Despite its advantages, AI also presents challenges such as ethical concerns, bias in AI models, and job displacement. It is crucial to implement responsible AI practices to ensure fairness, transparency, and accountability as AI continues to evolve.
Evaluation Metrics:
ROUGE-1 Score: 0.65
ROUGE-2 Score: 0.63
BLEU Score: 0.34
Developed by Nakul Limbani


AI-Powered Text Summarizer
Enter your text:
even predicting outbreaks. Meanwhile, in finance, AI enhances fraud detection, automates trading, and improves customer support through AI-powered chatbots. Despite its advantages, AI also presents challenges such as ethical concerns, bias in AI models, and job displacement. It is crucial to implement responsible AI practices to ensure fairness, transparency, and accountability as AI continues to evolve and integrate into society
Select Model:
TextRank (Extractive)
Summarize
Summary:
It is crucial to implement responsible AI practices to ensure fairness, transparency, and accountability as AI continues to evolve and integrate into society Meanwhile, in finance, AI enhances fraud detection, automates trading, and improves customer support through AI-powered chatbots. AI-powered systems are capable of processing vast amounts of data, identifying patterns, and making data-driven decisions.
Evaluation Metrics:
ROUGE-1 Score: 0.72
ROUGE-2 Score: 0.7
BLEU Score: 0.42
Developed by Nakul Limbani

7. Future Work

- Fine-tune models on domain-specific data (e.g., Finance, Healthcare, News).
- Improve UI with additional customization options.
- Implement multilingual summarization support.
- Deploy the app as a cloud-based API service.

8. Conclusion

This project successfully developed an AI-powered Text Summarization Tool that leverages pre-trained transformer models and extractive algorithms. The web-based application allows users to generate high-quality summaries and compare model performances. Future improvements include fine-tuning, multilingual support, and API deployment.